Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Davies, Peter Ph.D.		
Project Title:	Cardiovascular Genomics in Gravitational Transitions		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	215-573-6813
Organization Name:	University of Pennsylvania		
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PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19041	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	09/01/2001	**End Date:**	12/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	SMS.002.08		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	When changes in the biomechanical environment of the circulation occur, blood vessels undergo well-orchestrated structural and metabolic remodeling to restore optimal function. We propose that this remarkable adaptive ability lies at the center of orthostatic intolerance exhibited by most astronauts on return to earth?s gravitational field after modest-to-long periods in microgravity. We are therefore mapping gene expression (transcription profiling) of the different vascular steady states exhibited in vivo (mouse) in simulated hypergravity and microgravity, and the transitions between them, in order to design better countermeasures for undesired vascular consequences in long-term space flight. The central objective of this project is to develop in a well defined mouse model a reference databank of gene expression profiles of the normal mammalian vasculature as a function of gravitational history and gravitational transition using RNA amplification technology developed to detect quantitative differences in multiple gene expressions in theheart, vasculature and lungs. Refinement to small amounts of tissues, RNA amplification, and array technology will be used to study many genes simultaneously and a well-developed bioinformatics program will ensure efficient data		

	analysis, annotation, and dissemination. The program uses C57/BL6 mice as the model organism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fidelity analyses of expression profiling of very small amounts of tissues/cells is a developing technology of immense value for high-throughput site specific phenotyping. Our findings support the 'high risk' conviction that the techniques, essential for the gravity experiments, will provide a comprehensive profile of cardiovascular gene expression for biomedicine in general.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)